

An exceptional four-dimensional unitary Hecke Yang–Baxter operator

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Abstract

We construct an explicit 16×16 unitary Yang–Baxter operator in the class

$$\left[e^{i\pi/3}, \frac{1}{2}, 4 \right].$$

It is the smallest-dimensional unresolved member of the sole remaining existence family in Lechner’s classification up to braid-character equivalence, and it corresponds to the ordinary-localization question highlighted by Galindo, Hong, and Rowell for the $\mathcal{C}(\mathfrak{sl}_3, 6)$ Jones–Wenzl sequence. Writing $\mathbb{C}^4 \cong \mathbb{C}^2 \otimes \mathbb{C}^2$, the associated reflection has a five-word real Pauli expansion. Four words combine into an involution M , a fifth involution E anticommutes with M , and the braid relation selects

$$H = \sqrt{\frac{2}{3}} M - \frac{1}{\sqrt{3}} E$$

from the circle of involutions $\alpha M + \beta E$. The resulting spectral projection has both unnormalized partial traces equal to $2\mathbf{1}_4$, so normalized matrix trace induces the $\eta = 1/2$ Markov trace. Hence the tensor-space representation induces a faithful representation of $H_n(3, 6)$ for every n , providing an ordinary unitary localization of the $\mathcal{C}(\mathfrak{sl}_3, 6)$ Jones–Wenzl sequence. Together with the known dimension-two obstruction and a dimension-three argument, this proves that the minimum local dimension is four.

Keywords. Yang–Baxter operator; Hecke algebra; braid group representation; unitary localization; fusion category.

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1 Introduction and main result

Let V be a finite-dimensional complex Hilbert space. A unitary $R \in \text{End}(V \otimes V)$ is a Yang–Baxter operator when

$$(R \otimes \mathbf{1}_V)(\mathbf{1}_V \otimes R)(R \otimes \mathbf{1}_V) = (\mathbf{1}_V \otimes R)(R \otimes \mathbf{1}_V)(\mathbf{1}_V \otimes R). \quad (1)$$

Lechner [3] classified the possible characters of non-involutive unitary R -matrices with two eigenvalues, up to one exceptional family. In his normalization the two eigenvalues are -1 and q , the (-1) -spectral projection P has normalized trace

$$\eta = \frac{\text{rank } P}{(\dim V)^2},$$

and the unresolved classes are

$$\left[e^{i\pi/3}, \frac{1}{2}, 2m \right] \quad (m \geq 1).$$

The class in base dimension two is empty; base dimension four is the smallest-dimensional unresolved member of this remaining family.

This question traces back to the case study of Galindo, Hong, and Rowell [1, §5.6]. They showed that the Jones–Wenzl representations associated with $\mathcal{C}(\mathfrak{sl}_3, 6)$ have a unitary quasi-localization and an 8×8 $(3, 1)$ -generalized localization, but no two-dimensional ordinary unitary localization. They wrote that the sequence did not appear to have an ordinary localization and noted it as a possible counterexample to the localization conjecture. Lechner’s work later isolated the precise matrix family above.

The following is the central result of this paper.

Theorem 1.1. *Let*

$$q = e^{i\pi/3} = \frac{1 + i\sqrt{3}}{2}.$$

There is an explicit unitary $R \in \text{End}(\mathbb{C}^4 \otimes \mathbb{C}^4)$ such that

$$(R + \mathbf{1}_{16})(R - q\mathbf{1}_{16}) = 0, \quad R_1 R_2 R_1 = R_2 R_1 R_2,$$

and each eigenvalue $-1, q$ has multiplicity eight. Its (-1) -spectral projection P satisfies

$$\text{Tr}_1(P) = \text{Tr}_2(P) = \mathbf{21}_4.$$

Therefore

$$\left[e^{i\pi/3}, \frac{1}{2}, 4 \right] \neq \emptyset,$$

and the associated tensor-space representations induce faithful embeddings

$$H_n(3, 6) \hookrightarrow \text{End}((\mathbb{C}^4)^{\otimes n}) \quad (n \geq 2).$$

Under the identification in [1, §5.6], these embeddings form an ordinary unitary localization of the $\mathcal{C}(\mathfrak{sl}_3, 6)$ Jones–Wenzl representation sequence.

Corollary 1.2. *The minimum local dimension of an ordinary unitary localization of this Jones–Wenzl sequence is four.*

The formula for R is given in Section 2; the matrix identities are proved in Sections 3 and 4. Section 5 proves the full localization claim rather than only the two-site spectral statement. The minimality argument is in Section 6. Section 7 records a generalized Yang–Baxter normal form, and Section 9 separates the exact theorem from the necessarily incomplete priority audit.

2 The five-word construction

Put $V = \mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4$, and define the real matrices

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

A word such as $ZIZZ$ denotes $Z \otimes I \otimes Z \otimes Z$ on $V \otimes V$. Define

$$\begin{aligned} H = & -\frac{1}{\sqrt{6}}ZIZZ - \frac{1}{\sqrt{6}}ZIJJ - \frac{1}{\sqrt{6}}JIZJ + \frac{1}{\sqrt{6}}JIJZ \\ & - \frac{1}{\sqrt{3}}XIXX, \end{aligned} \quad (2)$$

and

$$R = \frac{q-1}{2}\mathbf{1}_{16} + \frac{q+1}{2}H = \frac{-1+i\sqrt{3}}{4}\mathbf{1}_{16} + \frac{3+i\sqrt{3}}{4}H. \quad (3)$$

Equations (2)–(3) are an exact specification of all 256 matrix entries in the computational basis.

The relevant multiplication table is

$$\begin{aligned} X^2 = Z^2 = I, \quad J^2 = -I, \\ XZ = J = -ZX, \quad XJ = Z = -JX, \quad JZ = X = -ZJ. \end{aligned}$$

Thus every product of tensor words reduces to a signed word over $\{I, X, Z, J\}$.

3 A circle of reflections

The five terms in (2) have a compact algebraic explanation. Set

$$A = ZIZZ, \quad B = ZIJJ, \quad C = JIZJ, \quad D = JIJZ, \quad E = XIXX$$

and

$$M = \frac{-A - B - C + D}{2}. \quad (4)$$

Lemma 3.1. *The operators A, B, C, D, E are real symmetric involutions. The first four commute, $D = ABC$, and E anticommutes with each of A, B, C, D . Consequently*

$$M^2 = E^2 = \mathbf{1}_{16}, \quad ME = -EM.$$

For real α, β ,

$$H(\alpha, \beta) = \alpha M + \beta E$$

satisfies

$$H(\alpha, \beta)^2 = (\alpha^2 + \beta^2)\mathbf{1}_{16}.$$

Proof. The word multiplication table gives every stated commutation relation and $D = ABC$. One may also prove $M^2 = \mathbf{1}$ without expanding. Simultaneously diagonalize A, B, C , and let their joint eigenvalues be $x, y, z \in \{\pm 1\}$. Since $D = ABC$, the eigenvalue of M is

$$\frac{-x - y - z + xyz}{2} \in \{\pm 1\}.$$

The anticommutation $ME = -EM$ then makes the cross term in $(\alpha M + \beta E)^2$ vanish. $\square$

Remark 3.2 (Graph-phase form). If $x = (-1)^a, y = (-1)^b, z = (-1)^c$, then

$$\frac{-x - y - z + xyz}{2} = -(-1)^{ab+ac+bc}.$$

Thus in the joint stabilizer eigenbasis, M is the negative phase associated with the triangle graph, while E flips all three syndrome bits. This explains the four-plus-one support in (2); no graph-state formalism is needed in the proof.

4 The cubic certificate and Yang–Baxter equation

For any $K \in \text{End}(V \otimes V)$, write

$$K_1 = K \otimes \mathbf{1}_V, \quad K_2 = \mathbf{1}_V \otimes K.$$

The Yang–Baxter equation for (3) will follow from a cubic identity for H .

Proposition 4.1. *On the unit circle $\alpha^2 + \beta^2 = 1$,*

$$H(\alpha, \beta)_1 H(\alpha, \beta)_2 H(\alpha, \beta)_1 - H(\alpha, \beta)_2 H(\alpha, \beta)_1 H(\alpha, \beta)_2 = \frac{1}{3} (H(\alpha, \beta)_1 - H(\alpha, \beta)_2) \quad (5)$$

if and only if

$$\beta^2 = \frac{1}{3}, \quad \alpha^2 = \frac{2}{3}.$$

In particular, the four signed points with these squared coordinates are exactly the solutions inside this reflection circle.

Proof. Reduce the left side of (5), minus its right side, in the six-letter word basis. Exactly the following 18 words survive:

coefficient	words
$-\alpha(3\beta^2 - 1)/6$	$I I J I J Z, J I Z J I I, Z I J J I I, Z I Z Z I I$
$+\alpha(3\beta^2 - 1)/6$	$I I J I Z J, I I Z I J J, I I Z I Z Z, J I J Z I I$
$-\beta(3\alpha^2 - 3\beta^2 - 1)/3$	$I I X I X X$
$+\beta(3\alpha^2 - 3\beta^2 - 1)/3$	$X I X X I I$
$+\alpha(\alpha^2 - 2\beta^2)/2$	$J I J Z X X, J I Z J X X, X I J X Z J, Z I Z Z X X$
$-\alpha(\alpha^2 - 2\beta^2)/2$	$X I J X J Z, X I Z X J J, X I Z X Z Z, Z I J J X X$

This is a finite certificate: each row follows by repeated use of the displayed 2×2 multiplication table. On $\alpha^2 + \beta^2 = 1$,

$$\alpha^2 - 2\beta^2 = 1 - 3\beta^2, \quad 3\alpha^2 - 3\beta^2 - 1 = 2(1 - 3\beta^2).$$

Hence all coefficients vanish when $3\beta^2 = 1$. Conversely, $\alpha = 0$ or $\beta = 0$ leaves a displayed coefficient nonzero, and when $\alpha\beta \neq 0$ the first and third groups force $3\beta^2 = 1$. The claim follows. $\square$

Choosing

$$\alpha = \sqrt{\frac{2}{3}}, \quad \beta = -\frac{1}{\sqrt{3}}$$

in $H(\alpha, \beta)$ gives exactly (2). Lemma 3.1 and Proposition 4.1 therefore give

$$H^* = H, \quad H^2 = \mathbf{1}_{16}, \quad H_1 H_2 H_1 - H_2 H_1 H_2 = \frac{1}{3} (H_1 - H_2). \quad (6)$$

Every word in (2) contains a traceless factor, so $\text{Tr } H = 0$. The eigenvalues $+1$ and -1 thus both have multiplicity eight.

Proof of the matrix part of Theorem 1.1. Put

$$a = \frac{q-1}{2}, \quad b = \frac{q+1}{2},$$

so $R = a\mathbf{1} + bH$. Since $H_i^2 = \mathbf{1}$, direct expansion gives

$$R_1 R_2 R_1 - R_2 R_1 R_2 = b [a^2(H_1 - H_2) + b^2(H_1 H_2 H_1 - H_2 H_1 H_2)].$$

But

$$\frac{a}{b} = \frac{q-1}{q+1} = \frac{i}{\sqrt{3}},$$

so $a^2 + b^2/3 = 0$. Equation (6) proves the Yang–Baxter equation.

On the $+1$ -eigenspace of H , $R = a + b = q$; on the -1 -eigenspace, $R = a - b = -1$. Therefore

$$(R + \mathbf{1})(R - q\mathbf{1}) = 0, \quad R^* R = \mathbf{1},$$

with both eigenvalues of multiplicity eight. The (-1) -spectral projection is

$$P = \frac{\mathbf{1}_{16} - H}{2},$$

so $P = P^* = P^2$, $\text{rank } P = 8$, and $\eta = 8/16 = 1/2$. Finally, each word in H has a traceless factor on each four-dimensional site. Hence

$$\text{Tr}_1(H) = \text{Tr}_2(H) = 0, \quad \text{Tr}_1(P) = \text{Tr}_2(P) = 2\mathbf{1}_4.$$

□

5 The full localization statement

We now verify that the construction localizes the whole Jones–Wenzl tower, not merely its two-site generator. Let $H_n(q)$ be the specialized Hecke algebra of type A_{n-1} , generated by $g_1, \dots, g_{n-1}$, with the braid and far-commutativity relations and

$$(g_i + \mathbf{1})(g_i - q\mathbf{1}) = 0.$$

Its idempotent generators e_i are related to the braid generators by

$$g_i = q\mathbf{1} - (1 + q)e_i.$$

The Yang–Baxter and Hecke relations define a representation

$$\rho_n : H_n(q) \longrightarrow \text{End}(V^{\otimes n}), \quad \rho_n(g_i) = R_i, \quad \rho_n(e_i) = P_i.$$

Let

$$\tau_n(T) = 4^{-n} \text{Tr}(T)$$

be normalized matrix trace.

Equip $H_n(q)$ with the conjugate-linear involution

$$e_i^* = e_i, \quad \text{equivalently} \quad g_i^* = g_i^{-1}.$$

Since $P_i = P_i^*$ and R_i is unitary, each ρ_n is a unital $*$ -representation. If $\iota_n : H_n(q) \rightarrow H_{n+1}(q)$ is the standard inclusion, then

$$\rho_{n+1}(\iota_n(x)) = \rho_n(x) \otimes \mathbf{1}_4. \tag{7}$$

For the $\eta = 1/2$ Markov trace, write

$$\text{Ann}_n = \{x \in H_n(q) : \text{tr}_{1/2}(yx) = 0 \text{ for every } y \in H_n(q)\}.$$

Proposition 5.1. *The traces $\tau_n \circ \rho_n$ are the Markov traces with parameter $\eta = 1/2$:*

$$\tau_{n+1}((\rho_n(x) \otimes \mathbf{1}_4)P_n) = \frac{1}{2} \tau_n(\rho_n(x)) \quad (x \in H_n(q)). \quad (8)$$

For every n ,

$$\ker \rho_n = \text{Ann}_n.$$

Consequently ρ_n induces an injective representation of the semisimple quotient

$$H_n(3, 6) = H_n(q) / \text{Ann}_n.$$

Proof. Taking the partial trace over the last tensor factor and using $\text{Tr}_2(P) = 2\mathbf{1}_4$ gives

$$\begin{aligned} \tau_{n+1}((\rho_n(x) \otimes \mathbf{1}_4)P_n) &= 4^{-(n+1)} \text{Tr}(\rho_n(x) \text{Tr}_{n+1}(P_n)) \\ &= 4^{-(n+1)} 2 \text{Tr}(\rho_n(x)) = \frac{1}{2} \tau_n(\rho_n(x)). \end{aligned}$$

Moreover,

$$\tau_{n+1}(\rho_n(x) \otimes \mathbf{1}_4) = \tau_n(\rho_n(x)),$$

so the traces are consistent under the standard inclusions. Together with normalization and traciality, this is the defining Markov property. Uniqueness of the Markov trace with prescribed $\text{tr}(e_i) = 1/2$ identifies it with the trace defining $H_n(3, 6)$ [6]. Explicitly, in the normalization used for this quotient,

$$\eta_{6,3} = \frac{1 - q^{-2}}{(1 + q)(1 - q^{-3})} = \frac{1}{2}, \quad q = e^{i\pi/3}.$$

This is the identification used in [1, Theorem 5.28].

If $x \in \text{Ann}_n$, take $y = x^*$ to obtain

$$0 = \text{tr}_{1/2}(x^*x) = \tau_n(\rho_n(x)^* \rho_n(x)).$$

Normalized matrix trace is faithful, hence $\rho_n(x) = 0$. The reverse inclusion is immediate. Thus the kernel is exactly the annihilator and the induced map on the quotient is injective. $\square$

Equation (7) and Proposition 5.1 also give

$$\iota_n^{-1}(\text{Ann}_{n+1}) = \text{Ann}_n.$$

Thus the inclusions descend to the semisimple quotients, and the induced injections $\bar{\rho}_n$ satisfy

$$\bar{\rho}_{n+1}(\bar{\iota}_n(x)) = \bar{\rho}_n(x) \otimes \mathbf{1}_4.$$

Galindo, Hong, and Rowell recall the braid-equivariant isomorphism

$$\text{End}_{\mathcal{C}}(X^{\otimes n}) \cong H_n(3, 6)$$

for the two-dimensional simple generator X of $\mathcal{C}(\mathfrak{sl}_3, 6)$ [1, §5.6]. Proposition 5.1 and the displayed tower compatibility therefore give a faithful unitary $*$ -morphism of towers, completing the localization assertion in Theorem 1.1 in the sense of [1, Definition 4.10].

Corollary 5.2. *This Jones–Wenzl sequence is not the possible counterexample to the localization conjecture contemplated in [1, Remark 6.2(2)].*

Proof. Theorem 1.1 supplies the ordinary unitary localization whose possible nonexistence motivated that remark. $\square$

6 Minimality and amplification

Proof of Corollary 1.2. A one-dimensional local space cannot carry the required two-eigenvalue generator. Galindo, Hong, and Rowell prove that no two-dimensional ordinary unitary localization exists [1, Lemma 5.26].

It remains to exclude local dimension three. Suppose such an ordinary unitary localization existed, with local braid operator $F \in \text{End}(\mathbb{C}^3 \otimes \mathbb{C}^3)$. Since a localization intertwines the braid generators,

$$(F + \mathbf{1}_9)(F - q\mathbf{1}_9) = 0.$$

Injectivity at two strands rules out either spectral projection vanishing, so both eigenvalues -1 and q occur. Thus F belongs to one of the dimension-three Hecke classes in [3, Theorem 3.4]. At $q = e^{i\pi/3}$, the possible Markov parameters are $1/3, 1/2, 2/3$. The value $1/2$ is impossible in dimension three because it would require a projection of rank $9/2$. At $1/3$ the representation is of Temperley–Lieb type; at $2/3$ its complementary projection is of Temperley–Lieb type [3, Lemma 3.5 and the following discussion].

Both cases kill an element that is nonzero in the exceptional quotient. Indeed, in $H_3(3, 6)$ put

$$e_i^\perp = \mathbf{1} - e_i, \quad c = \frac{q}{(1+q)^2} = \frac{1}{3},$$

$$T = e_1 e_2 e_1 - c e_1, \quad T^\perp = e_1^\perp e_2^\perp e_1^\perp - c e_1^\perp.$$

The calculation in [3, Lemma 3.5], applied to e_i and to the flipped Hecke projections $e_i^\perp$, gives

$$\text{tr}_{1/2}(T^*T) = \text{tr}_{1/2}((T^\perp)^*T^\perp) = \frac{1}{18} > 0.$$

Hence both elements are nonzero in $H_3(3, 6)$. A parameter- $1/3$ candidate kills T , while a parameter- $2/3$ candidate kills $T^\perp$, so neither can be faithful. Theorem 1.1 supplies dimension four. $\square$

There is also an immediate infinite family. If $\mathbf{1}_m$ denotes the identity Yang–Baxter operator on $\mathbb{C}^m \otimes \mathbb{C}^m$, the reshuffled tensor product $R \boxtimes \mathbf{1}_m$ is a unitary Yang–Baxter operator on a base space of dimension $4m$ [3, §3]. Its two eigenvalues and normalized spectral rank are unchanged. Therefore

$$\left[e^{i\pi/3}, \frac{1}{2}, 4m \right] \neq \emptyset \quad (m \geq 1). \quad (9)$$

This does not settle the remaining base dimensions $6, 10, 14, \dots$

Corollary 6.1. *The Yang–Baxter endomorphism associated with R , in the terminology used by Lechner, has Jones index 4 and is standard braided.*

Proof. Lechner observes that every member of the exceptional class $[e^{i\pi/3}, 1/2, 2m]$ has these two properties [3, p. 16]; Theorem 1.1 supplies such a member. $\square$

7 A (3,2)-generalized normal form

The visible identity in the second qubit of every word in (2) does have a precise chain-level interpretation, but a coordinate swap is essential. Write the i -th four-dimensional site as

$$V_i = \mathbb{C}_{a_i}^2 \otimes \mathbb{C}_{b_i}^2,$$

let $\Sigma(u \otimes v) = v \otimes u$ be the within-site qubit swap, and put

$$U_n = \Sigma^{\otimes n}.$$

In the factor order obtained after applying U_n ,

$$(b_1, a_1, b_2, a_2, \dots),$$

define

$$K = \frac{q-1}{2} \mathbf{1}_8 + \frac{q+1}{2} K_H$$

and

$$K_H = -\frac{ZZZ + ZJJ + JJZ - JZJ}{\sqrt{6}} - \frac{XXX}{\sqrt{3}}. \quad (10)$$

Let $\rho_R^{(n)}$ denote the standard braid representation generated by the adjacent copies of R .

Proposition 7.1. *The two-site operator factors after the sitewise swap as*

$$U_2 R U_2^* = \mathbf{1}_{b_1} \otimes K_{a_1, b_2, a_2}.$$

The operator $K \in \text{End}((\mathbb{C}^2)^{\otimes 3})$ is unitary and satisfies the $(3, 2)$ -generalized Yang–Baxter equation

$$(K \otimes \mathbf{1}_4)(\mathbf{1}_4 \otimes K)(K \otimes \mathbf{1}_4) = (\mathbf{1}_4 \otimes K)(K \otimes \mathbf{1}_4)(\mathbf{1}_4 \otimes K) \quad (11)$$

on $(\mathbb{C}^2)^{\otimes 5}$, together with far commutativity. If $\rho_K^{(n)}$ denotes the resulting $(3, 2)$ -generalized braid representation, explicitly

$$\rho_K^{(n)}(\sigma_i) = \mathbf{1}_2^{\otimes 2(i-1)} \otimes K \otimes \mathbf{1}_2^{\otimes 2(n-i-1)},$$

then for every n and every braid-algebra element x ,

$$U_n \rho_R^{(n)}(x) U_n^* = \mathbf{1}_{b_1} \otimes \rho_K^{(n)}(x).$$

Consequently K gives a faithful unitary $(3, 2)$ -localization of $H_n(3, 6)$ in the convention of [1, Definition 4.24].

Proof. Conjugating the five words in (2) by U_2 puts the identity factor first and leaves exactly (10) on (a_1, b_2, a_2) . This proves the factorization and unitarity. On a chain, adjacent copies of K act on triples shifted by two qubits. After factoring out the global spectator b_1 , the ordinary Yang–Baxter equation (1) is precisely (11); nonadjacent triples are disjoint, so far commutativity holds. The displayed chain conjugacy follows generator by generator. Tensoring by a spectator identity does not change the kernel, and Proposition 5.1 proves faithfulness on $H_n(3, 6)$. $\square$

Equivalently, any $(3, 2)$ -generalized operator K on $W^{\otimes 3}$ blocks into an ordinary operator $\mathbf{1}_W \otimes K$ on the local space $W \otimes W$, in the displayed ordering; the ordinary braid relation is $\mathbf{1}_W$ tensored with (11). This elementary blocking observation explains why the present 8×8 active operator yields a 16×16 ordinary one.

The known operator in [1, Eq. (5.2), Theorem 5.28, and Remark 5.29(1)] is instead of type $(3, 1)$ and comes from a quaternionic realization. No equivalence between that generalized localization and (10) is claimed here. The same paper also records a different $(3, 2)$ operator in Example 3.14. Its two eigenvalues have projective ratio i (up to inversion), whereas the ratio for K is $-q = e^{-2\pi i/3}$ (up to inversion). The two local operators are therefore not unitarily conjugate up to a phase. In particular, being of type $(3, 2)$ is not by itself the new feature here; the exceptional Hecke spectrum and faithful $(3, 6)$ localization are.

8 Verification and provenance

The public source package contains three supported verification routes and the unaltered original attachment from which one route was hardened.

1. `verify_supplied.py` is a hardened SymPy implementation of the original attachment. It uses explicit, optimization-safe checks and reconstructs the matrices and main identities exactly. The byte-for-byte original is retained separately as `verify_supplied_original.py` for provenance and is not a supported execution path.
2. `verify_exact.py` is an independent implementation using only the Python standard library. It performs sparse exact arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$, directly checks the 16×16 identities, the 64×64 Yang–Baxter equation, both partial traces, both dimension-three obstruction norms, the active Hecke relation, the literal printed active operator, the 32×32 generalized equation (11), and far commutativity.
3. `verify_tensor_words.py` never constructs a matrix. It reduces the abstract Pauli-word algebra and checks the complete 18-word certificate in Proposition 4.1.

The latter two programs were written independently of the supplied checker. All comparisons are exact; floating-point tolerances are not used. These finite computations verify the displayed matrix, word, trace, and obstruction identities. Tower faithfulness, the dimension-three classification step, and the priority conclusion remain mathematical or literature arguments rather than finite computational checks.

The five-word support was originally found by an AI-assisted structured numerical search in a real Pauli–Clifford basis. The discovery code and random seeds were not retained. Accordingly, this paper makes no claim that the search was reproducible, exhaustive, or established uniqueness. The generative-AI systems identified in the declaration below were also used for mathematical exploration and derivation, adversarial proof analysis, verifier development, literature support, and manuscript preparation. Every claim in the theorem is supported by the printed derivation, and the finite identities and trace calculations are checked by the exact programs described above. No mathematical conclusion depends on accepting an AI-generated assertion or on reproducing the original numerical search.

9 Related work, novelty, and limitations

Galindo, Hong, and Rowell’s 2012 advance publication studied the $\mathcal{C}(\mathfrak{sl}_3, 6)$ sequence, excluded ordinary local dimension two, and supplied quasi- and $(3, 1)$ -generalized localizations, but no ordinary dimension-four operator [1, §5.6]. Rowell’s quaternionic model and the localization and unitary-braided-vector-space literature provide close prior art [4, 5, 2]. Lechner then identified $[e^{i\pi/3}, 1/2, 2m]$ as the sole remaining existence family up to braid-character equivalence and stated that its dimension-two member is empty [3, p. 16].

A focused search through 16 August 2026 of primary sources, scholarly indexes, general web results, later papers mentioning the preprint, and public code found no earlier formula equivalent to (2) and no published ordinary four-dimensional localization. The defensible conclusion is therefore that the construction *appears new* and answers the open class identified in [3]. This is not proof of absolute priority: an unindexed, unpublished, paywalled, differently parameterized, or concurrent equivalent may exist.

The broad ordinary-localization question dates to the 2012 case study; the exact 16×16 formulation was isolated in 2026. The result does not classify the exceptional family in every even dimension, prove uniqueness, or identify an intertwiner with the known quaternionic $(3, 1)$ model.

Data, code, and source availability

The source, exact verifiers, frozen outputs, and audit materials are maintained in the [public source package](#) in the [AlecKriebel/Math repository](#). The curated submission package supplies SHA-256 checksums and explicit manuscript and code licenses. The maintained [project page](#) links the current edition. No empirical dataset was generated or analyzed; the exact code, frozen outputs, and sources reproduce all computational claims and accompany this manuscript in a dedicated archival record whose DOI is cited in the submitted edition.

CRedit authorship contribution statement

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OpenAI ChatGPT 5.6 Sol Pro through chatgpt.com and OpenAI ChatGPT 5.6 Sol Ultra through the ChatGPT desktop application were used for research, review, and composition. Anthropic Claude Opus 5 Max through claude.ai was used for review. Uses included mathematical exploration and derivation, adversarial proof analysis, symbolic and software work, literature support, and manuscript preparation. The work was developed through an iterative, human-directed research process. The author determined the scope and claims released here, curated the resulting package, reviewed and edited the content, and assumes responsibility for the submitted manuscript and public verification materials. The author takes full responsibility for every claim released here. The proof and exact artifacts are supplied to make the AI-mediated research independently checkable.

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